Question	Answer	Marks	Partial Marks
3(a)	6.06 or 6.060 to 6.061	**3**	**M2** for $\frac{82500-77500}{82500}[\times 100]$ oe or **M1** for $\frac{77500}{82500}[\times 100]$ soi
3(b)	13 674 cao	**3**	**M1** for $12000\left(1+\frac{2.2}{100}\right)^6$ **A1** for 13673.7...
4(a)(i)	Translation $\begin{pmatrix}-8\\2\end{pmatrix}$ oe	**2**	**B1** for each
4(a)(ii)	Enlargement [sf =] $\frac{1}{2}$ oe $(-4, 0)$	**3**	**B1** for each
4(a)(iii)	Rotation 90° clockwise oe $(1, -1)$	**3**	**B1** for each
4(b)	Triangle with (1, –1), (5, –1), (1,7)	**2**	**B1** for correct size and orientation in wrong position or for 3 correct points not joined
5(a)(i)	$(2n+m)(m-3)$ final answer	**2**	**M1** for $m(2n+m)-3(2n+m)$ or $2n(m-3)+m(m-3)$
5(a)(ii)	$(2y-9)(2y+9)$ final answer	**1**	
5(a)(iii)	$(t-4)(t-2)$ final answer	**2**	**B1** for $(t-4)(t-2)$ seen and spoiled or **M1** for $t(t-2)-4(t-2)$ or $t(t-4)-2(t-4)$ or $(t+a)(t+b)$ where $a+b=-6$ or $ab=+8$
5(b)	$[x=]\frac{2m}{k+1}$	**4**	**M1** for $xk=2m-x$ or $k=\frac{2m}{x}-1$ **M1** for $xk+x=2m$ or $k+1=\frac{2m}{x}$ **M1** for $x(k+1)=2m$

Question	Answer	Marks	Partial Marks
5(c)	correctly eliminating one variable	**M1**	
	[x =] 6	**A1**	
	[y =] –2	**A1**	If 0 scored **SC1** for 2 values satisfying one of the original equations or **SC1** if no working shown, but 2 correct answers given
5(d)(i)	$3m-4(m+4)=6m(m+4)$	**M1**	or $\frac{3m-4(m+4)}{m(m+4)}[=6]$ oe
	$3m-4m-16=6m^2+24m$	**M1**	removes brackets correctly
	$6m^2+25m+16=0$	**A1**	with no errors or omissions
5(d)(ii)	$\frac{-25\pm\sqrt{(25)^2-4(6)(16)}}{2\times 6}$ or $\frac{-25}{12}\pm\sqrt{\left(\frac{25}{12}\right)^2-\frac{16}{6}}$	**2**	**B1** for $\sqrt{(25)^2-4(6)(16)}$ or better or **B1** for $\left(m+\frac{25}{12}\right)^2$ and if in form $\frac{p+\sqrt{q}}{r}$ or $\frac{p-\sqrt{q}}{r}$ **B1** for $p=-25$ and $r=2(6)$
	–0.79 and –3.38 final ans cao	**2**	**B1** for each **SC1** for –0.8 and –3.4 or for –0.78 and –3.37 or –0.789... and –3.377... or 0.79 and 3.38 or –0.79 and –3.38 seen in working
6(a)	4.79 or 4.788 to 4.789	**3**	**M2** for $\sqrt[3]{\frac{230\times 3}{2\times\pi}}$ oe or **M1** for $230=\frac{2}{3}\times\pi\times r^3$ oe If 0 scored **SC1** for answer 3.8[0…]
6(b)(i)	8.7[0] or 8.702 to 8.704	**3**	**M2** for $(300-230)\div(1.6^2\pi)$ or **M1** for $\pi\times 1.6^2\times h$
6(b)(ii)	6.4	**3**	**M2** for $1.6\times\sqrt[3]{\frac{19200}{300}}$ oe or **M1** for sf $\sqrt[3]{\frac{19200}{300}}$ or $\sqrt[3]{\frac{300}{19200}}$ oe or for $\left(\frac{1.6}{r}\right)^3=\frac{300}{19200}$